LIS-SRS-001 · VERSION 1.3 · DRAFT FOR REVIEW · 2026-10-20

# Sky LIS — Business Software Requirements Specification

| | | | |
|---|---|---|---|
| **Prepared for** | National Technology — LIS project | **Prepared by** | Claude Code (AI analyst), for review by Mohamed Hashem |
| **Classification** | Internal — Confidential | **Basis** | Client Portal prototype v2 (full scope) and v2.1 (release-1 scope) · Super Admin Portal prototype v3 (full scope) and v3.1 (release-1 scope) · LIS Requirements Specification LIS-RS-001 · Prototype comparison LIS-PC-001 and build plan LIS-BP-001 · LDM 9.2/9.3 knowledge base (domain behaviour) |

**2** Portals

**16** Modules

**68** Client pages (28 in R1)

**21** Admin pages (17 in R1)

**12** Workflows

**28** Business rules

| Version | Date | Author | Change |
|---|---|---|---|
| 1.0 | 2026-10-09 | Claude Code | First complete business SRS covering both portals, all modules, pages and workflows, with the release-1 scope matrix. |
| 1.1 | 2026-10-14 | Claude Code | CP-41 Tests: parent services on the test page (the services carrying the test; join or leave profiles and cultures from the test, the other types set up from the service), as the Super Admin prototype shows. Page ids unchanged. |
| 1.2 | 2026-10-20 | Claude Code | Treasury removed from the system (change request LIS-CR-004, business owner decision of 2026-10-20): CP-28 Treasuries and CP-29 Treasury requests withdrawn; BR-27 withdrawn; CP-23 / CP-06 payments stamped with cashier, branch, method and reference only; CP-30 cash closing per cashier and branch; WF-09 without the treasury step; Payments back in the Finance menu; data model, status models, glossary and the security / audit summaries updated. Page and rule ids stay reserved. |
| 1.3 | 2026-10-20 | Claude Code | Sign-in by address (change request LIS-CR-005, business owner decision of 2026-10-20): every laboratory signs in at its own address <base>/lab/<slug> and the Super Admin at <base>/admin/login; the bare address asks for the laboratory address; CP-03 and SA-01 updated; new rule BR-28. |

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# 1. Introduction

## 1.1 Purpose

This document specifies, in business terms, what the Sky LIS product does: the two portals, their modules and pages, the data each page handles, the actions users can take, the rules the system enforces, and the end-to-end workflows that connect the pages. It is the reference for product owners, laboratory domain experts, designers, developers and testers, and the baseline for acceptance.

## 1.2 Scope

Sky LIS is a cloud, subscription-based, multi-tenant Laboratory Information System operated by National Technology. The Super Admin Portal serves National Technology staff; the Client Portal serves each subscribed laboratory (tenant) with its branches and staff. This SRS describes the full target product and marks the subset delivered in Release 1.

**Out of scope:**

- Installation, hosting and infrastructure procedures.
- Detailed visual design (the prototypes are the UI reference).
- Quality control, outsourcing to external laboratories, SMS campaigns and inter-branch transfer modules beyond the placeholders noted here.
- Country-specific regulatory reporting.

## 1.3 Conventions

| Convention | Meaning |
|---|---|
| Page ID | CP-nn for Client Portal pages, SA-nn for Super Admin pages. IDs are stable. |
| Release | R1 = included in Release 1 (prototypes v2.1 / v3.1). R2 = later release (full-scope prototypes v2 / v3). |
| Business rule | BR-nn, listed in section 9 and referenced from pages and workflows. |
| Workflow | WF-nn, step tables with actor, action and system response. |
| Requirement reference | Codes such as REG-005 refer to the functional requirements in LIS-RS-001. |

# 2. Business overview

## 2.1 Business context

National Technology replaces its on-premise LDM laboratory system with Sky LIS, a cloud product sold by subscription. A laboratory subscribes online, receives its own portal address and an independent copy of the global laboratory master data, and runs its daily operation — reception, samples, results, printing, money — from the Client Portal across its branches. National Technology operates the platform from the Super Admin Portal: client lifecycle, plans and wallets, global master data, libraries and platform configuration.

## 2.2 Business goals

1. Let a laboratory start operating within a day of subscribing, with a ready catalogue and price lists.
2. Preserve every operational safeguard laboratories rely on: duplicate-patient protection, the sample Received gate, verified-only printing, delta and panic alerts, contract-correct pricing, auditable money movements.
3. Serve reception, bench, sign-off, accounts and administration personas from one bilingual, responsive workspace.
4. Connect analysers and PCR instruments and release normal results automatically where the laboratory chooses to.
5. Give National Technology central control of master data, subscriptions and support.

## 2.3 Actors

| Actor | Responsibilities |
|---|---|
| Super Admin / Operations Admin | National Technology staff: clients, plans and wallets, admin users and roles, global master data, libraries, platform parameters, sessions and audit. |
| Lab Administrator | Owns the tenant configuration: branches, roles, users, catalogue copy, payers and prices, payment methods, parameters, analysers and automation. |
| Reception clerk / cashier | Registers patients and visits, orders services, answers preparation questions, collects money, prints labels, job orders, receipts and verified reports, closes the drawer. |
| Phlebotomist / courier | Collects samples in the laboratory or at home, is recorded as collector, brings lab-to-lab samples. |
| Technician / chemist | Receives samples, runs the worklist, enters or imports results, reviews, rejects unfit samples. |
| Unit head / responsible doctor | Verifies (e-signs) results, handles abnormal, panic and delta-flagged results, releases reports, owns auto-verification rules. |
| Microbiologist / pathologist | Enters culture results and formatted-text reports. |
| Accounts officer | Price lists and contracts, refunds and transfers, claims, financial reports. |
| Integration engineer | Analyser definitions and mappings, site agent, PCR kits. |
| Patient / payer / referring doctor | External parties receiving quotes, receipts, reports and claims. |

## 2.4 Key identifiers

| Identifier | Meaning |
|---|---|
| Patient No. | Permanent file number, auto-generated, never changes (P0004912). |
| Accession | Visit number: date-based sequence per tenant (2610080001). |
| Sample ID | Unique barcode number per tube, never reused (S2610080001). |
| Booking No. / Batch No. | Home-visit booking and batch (batch / lab-to-lab) grouping numbers. |
| Print list No. / Claim No. / Receipt No. | Numbers printed on reports, claims and receipts. |

## 2.5 Release scope summary

Release 1 delivers the cross-cutting platform plus every page that existed in the first prototypes with its enhancements (Client Portal v2.1) and the complete operator portal except the integration library (Super Admin v3.1). Release 2 adds the new client pages and the integration, automation and quality-control modules. In Release 1 registration omits the hospital Location/Department and Room fields and captures the payment before saving; samples are received automatically on save because the sample pages arrive in Release 2.

# 3. System overview

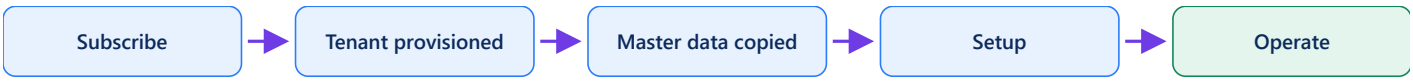
## 3.1 Two portals, one platform

**Super Admin Portal**

National Technology operates the platform: clients, plans and wallets, admin users and roles, global master data, integration libraries, platform parameters, sessions and audit.

**Client Portal**

Each laboratory runs reception, samples, laboratory, finance, setup, integration, automation and account from its own portal URL, across its branches, in Arabic or English.




Global master data flows from the Super Admin Portal into each tenant at creation (BR-25).

## 3.2 Cross-cutting capabilities

| Capability | Description |
|---|---|
| Tenancy & branches | Every record belongs to one tenant; users work in one branch per session; all transactions are branch-stamped. Global master data is copied to the tenant at creation and maintained independently afterwards (BR-25). |
| Bilingual, RTL, responsive | Arabic and English on every screen with true right-to-left layout; master data carries Arabic names; documents print in the configured language; desktop, tablet and phone layouts. |
| Security | Username/password, OTP for recovery and credential changes, password policy, session timeout, lock and expiry, roles with per-screen Query/Insert/Update/Delete, special rights, result rights per clinical group and branch. |
| Audit & notifications | Every create, update, delete, status change, print, payment, refund, transfer, cancellation and privilege change is logged with user, time, branch and device; a notification centre surfaces panic values, delays and agent status. |
| Grids & editors | Search, status and type filters, Excel export, pagination, row actions; editors validate mandatory fields, block deletion of referenced records and prefer deactivation. |
| Documents | Barcode labels (Code 128), job order, receipt, invoice, refund receipt, price quote, laboratory report with template, verifier signature and QR, abbreviated worksheet, worklist, cash-closing sheet, claim, analyser code sheet. |
| Keyboard | Ctrl+S saves the open form, Ctrl+K focuses search, F2 opens the service picker, Enter adds a typed service or shortcut, Esc closes dialogs. |

## 3.3 Modules

### Client Portal

| Module | Description | Release |
|---|---|---|
| **Access & onboarding** | Public website, self-service subscription, login, password recovery, branch selection. | R1 |
| **Overview** | Operational dashboard for the branch. | R1 |
| **Reception** | Patient files and visits in four registration modes, search, quotes, printing of verified reports. | R1 (walk-in, search, patients, quote) · R2 (home, batch, lab-to-lab, print reports) |
| **Samples** | Tube lifecycle: collection, receipt, rejection, replacement, tracking. | R2 |
| **Laboratory** | Worklist, result entry and sign-off, status work queue, quality control placeholder. | R1 (worklist, results by accession) · R2 (results by sample, work queue, QC) |
| **Finance** | Payers and pricing, payment methods, payments and money movements, closing, claims and reports. | R1 (payments, payment methods, payers, base price lists) · R2 (rest) |
| **Setup** | Tenant configuration and catalogue. | R1 (20 pages) · R2 (catalogue depth, parameters) |
| **Integration** | Analysers, code mapping, site agent, PCR kits. | R2 |
| **Automation** | Auto-verification settings and medical rules. | R2 |
| **Account** | Profile, wallet, support, audit log. | R1 (profile) · R2 (wallet, support, audit) |

### Super Admin Portal

| Module | Description | Release |
|---|---|---|
| **Overview** | Home and platform dashboard. | R1 |
| **Administration** | Clients, subscriptions and wallets, roles, admin users, sessions, audit log. | R1 |
| **Master data** | Global laboratory catalogue and lists copied to tenants. | R1 |
| **Microbiology** | Drug types, commercial names, antibiotics, organisms, sensitivity values. | R1 |
| **Integration library** | Analyser models, PCR kits, code systems, rule templates. | R2 |
| **Platform** | Platform parameters, notification templates, administrator profile. | R1 |

# 4. Client Portal — pages

> Each page card states its purpose, who uses it, the data it handles, the actions available, the business rules it enforces, the pages or master data it depends on, and its release.

## 4.1 Access & onboarding (4)

**CP-01** **Public website** Access & onboarding R1

Presents the product and routes visitors to login or subscription.

| | |
|---|---|
| Actors | Prospective laboratory |
| Key data | Product features, demo KPIs. |
| Actions | Login; Subscribe now; language switch. |
| Business rules | None. |
| Depends on | — |

**CP-02** **Subscription wizard** Access & onboarding R1

Creates a new tenant in four steps: lab details, OTP to the manager mobile, administrator credentials, portal URL delivery.

| | |
|---|---|
| Actors | Laboratory manager |
| Key data | Lab name (EN/AR), country, governorate, address, mobile, medical licence, manager name and mobile, email; username, password. |
| Actions | Send OTP; Verify; Create account; Login. |
| Business rules | Mobile validated by country length; password policy (8+, upper, lower, digit, special, match); OTP valid 5 minutes; tenant receives an independent copy of global master data (BR-25). |
| Depends on | Countries & governorates (SA) Notification templates (SA) |

**CP-03** **Login & password recovery** Access & onboarding R1

Authenticates the users of one laboratory at the laboratory address <base>/lab/<slug> (its name and logo shown) and recovers credentials with OTP; the bare address asks for the laboratory address.

| | |
|---|---|
| Actors | Laboratory users |
| Key data | Laboratory address (from the URL); username, password; username or mobile for recovery. |
| Actions | Login; Forgot password → OTP → new credentials; show/hide password; Find your laboratory (bare address). |
| Business rules | Only members of the laboratory of the address sign in there (BR-28); an unknown address shows a not-found state; lockout after repeated failures; locked or expired users refused; credentials updated only after OTP. |
| Depends on | Users, Notification gateway |

**CP-04** **Branch selection** Access & onboarding R1

Binds the session to one of the user's branches.

| | |
|---|---|
| Actors | All users |
| Key data | Assigned branches. |
| Actions | Continue. |
| Business rules | Only assigned branches listed; every transaction of the session is stamped with the branch. |
| Depends on | Branches, Users |

## 4.2 Overview (1)

**CP-05** **Dashboard** Overview R1

Shows the branch's day at a glance and offers quick actions.

| | |
|---|---|
| Actors | All users |
| Key data | KPIs: registrations, collected today, samples received*, pending results, unverified backlog, delayed tests, rejected samples*, wallet; 7-day registrations chart; delayed-tests table; recent activity. (*R2) |
| Actions | Quick actions to the main pages; refresh. |
| Business rules | Figures are computed for the session branch and day; delays derive from test durations (BR-13). |
| Depends on | Results, Samples, Audit log |

## 4.3 Reception (8)

**CP-06** **Registration** Reception R1 without Location/Department and Room; payment captured before saving

Registers a walk-in patient and visit: patient file, visit data, ordered services, preparation answers, discount, samples and documents; in Release 1 also the payment.

| | |
|---|---|
| Actors | Reception clerk |
| Key data | Patient: auto number, title, name, gender, date of birth / age, condition, mobile, national ID, nationality, email, governorate, city, address, comments, VIP. Visit: branch, location type, (R2: location/department, room), treatment no., referring doctor, payer, contract tier, insurance ID, diagnosis, urgent, biohazard, collected outside. Lines: service, status Ordered/Reserved, expected result, patient share, payer share, discount, collector. Preparation answers. Discount amount/% and reason. R1: payment method, amount, reference. |
| Actions | Search existing patient; new patient; add services by name, alias, short code, shortcut number, shortcut set or picker; set Reserved; remove line; apply discount; answer preparations; R1: choose payment method and amount; Save (Ctrl+S); edit a saved visit. |
| Business rules | Similar-name warning compares name and mobile (BR-01, REG-002); title sets gender; age ↔ DOB; gender-restricted services refused; frequency check alerts or blocks repeats (BR-12); contract alert conditions shown; auto-add rules add companion services; discount within the user cap with mandatory reason; tubes generated per group + sample type + condition (BR-02); Reserved keeps the line in the bill without a sample (BR-05); payer and tier set every price (BR-06); R1: paid amount ≤ amount due, reference mandatory when the payment method requires it; save prints labels and job order and the receipt per parameters. |
| Depends on | Services & tests, Payers & contracts, Contract pricing, Patient conditions, Preparations, Diagnoses, Shortcut sets, Test rules, Doctors, Reasons, Parameters, Payment methods |

**CP-07** **Home reservation** Reception R2

Books a home visit and converts it into a registration when the sample returns.

| | |
|---|---|
| Actors | Reception clerk, phlebotomist |
| Key data | Booking no., patient data, mandatory address, visit time, assigned phlebotomist, payer, services, quote, notes, status Booked/Converted/Cancelled. |
| Actions | New booking; preliminary labels and job order; cancel; convert to visit. |
| Business rules | Address mandatory; conversion creates the visit without retyping and marks the booking Converted. |
| Depends on | Registration, Users (phlebotomists), Services |

**CP-08** **Batch registration** Reception R2

Registers a group of people on one contract with the same services.

| | |
|---|---|
| Actors | Reception clerk |
| Key data | Company/group, payer and tier, doctor; member rows (patient no., title, name, gender, age, condition); service set; batch no. |
| Actions | Add row; look up patient number; choose services once; Save batch; go to batch payment. |
| Business rules | One accession per member under one batch number; duplicate check per row; documents printed for all members. |
| Depends on | Registration, Payers, Batch payment |

**CP-09** **Lab-to-Lab registration** Reception R2

Registers samples arriving from a contracted laboratory.

| | |
|---|---|
| Actors | Reception clerk |
| Key data | Sending lab (fills payer, tier, courier, phone), rows with patient data and services, collected-inside flag, batch no. |
| Actions | Choose lab; add rows; pick services per row; Save batch. |
| Business rules | Unticked "collected inside" marks the tube drawn outside and prints it on the job order; billing goes to the sending lab through claims. |
| Depends on | Sending labs, Registration, Claims |

**CP-10** **Search registrations** Reception R1

Finds visits and acts on them.

| | |
|---|---|
| Actors | Reception clerk, cashier |
| Key data | Accession, patient no., name, mobile, date range, branch, payer, status; row: services with reserved/cancelled styling, patient share, remaining balance. |
| Actions | Edit visit; samples (dialog in R1, page in R2); payments; documents (receipt, job order, labels, report); results; cancel services with reason. |
| Business rules | Verified services cannot be cancelled; cancellation keeps the record with reason and user (BR-15). |
| Depends on | Registration, Payments, Documents, Reasons |

**CP-11** **Patients** Reception R1

Patient files, history and documents.

| | |
|---|---|
| Actors | Reception clerk, lab administrator |
| Key data | Patient no., name, gender, age, mobile, condition, VIP, visit count, last visit; documents. |
| Actions | Search by number, name, mobile or national ID; visit history with results; attach documents; merge duplicates (right); new visit. |
| Business rules | Merge moves visits, results and payments to the surviving file and is audited; documents visible from every branch. |
| Depends on | Registration, Results, Audit log |

**CP-12** **Price query** Reception R1

Answers "how much, when, and must I fast?" without registering.

| | |
|---|---|
| Actors | Reception clerk |
| Key data | Payer, tier, services; list price, patient share, payer share, expected result, preparation. |
| Actions | Add services; print quote; continue to registration. |
| Business rules | Nothing is saved until the user continues to registration; prices come from contract pricing (BR-07). |
| Depends on | Contract pricing, Services |

**CP-13** **Print reports** Reception R2

Prints the day's verified reports in bulk.

| | |
|---|---|
| Actors | Reception clerk |
| Key data | Verified-unprinted report units, print list, print list no. |
| Actions | Add rows or whole accessions to the list; print list; abbreviated worksheet. |
| Business rules | Only verified reports print (BR-09); each print list gets a number printed on the reports. |
| Depends on | Work queue, Results |

## 4.4 Samples (4)

**CP-14** **Collection & receipt** Samples R2 in Release 1 samples are received automatically on save

Moves tubes through Ordered → Collected → Received by barcode.

| | |
|---|---|
| Actors | Phlebotomist, technician |
| Key data | Sample ID, accession, patient, unit, type, condition, services, status, events. |
| Actions | Scan in collect or receive mode; collect/receive all samples of an accession; today's list. |
| Business rules | Received gate: results cannot be entered before receipt (BR-03); auto-collect/receive parameters emulate the gate for non-scanning laboratories. |
| Depends on | Registration (samples) Parameters |

**CP-15** **Samples by visit** Samples R2

Lists a visit's tubes and reprints documents.

| | |
|---|---|
| Actors | Reception, technician |
| Key data | Per tube: unit, type, condition, status, status date, ID, labels, services. |
| Actions | Print one label; print all labels; job order; track. |
| Business rules | — |
| Depends on | Registration |

**CP-16** **Reject / unreject** Samples R2

Rejects unfit tubes and reinstates or replaces them.

| | |
|---|---|
| Actors | Technician |
| Key data | Sample ID, reason, accession, patient; rejected list. |
| Actions | Look up; reject with reason; unreject (same barcode); issue replacement (new barcode). |
| Business rules | Rejection needs a reason and notifies reception; replacement follows the tenant barcode policy (BR-18); rejected tubes accept no results. |
| Depends on | Reasons Labels Parameters |

**CP-17** **Sample tracking** Samples R2

Shows where a tube was and when.

| | |
|---|---|
| Actors | Everyone |
| Key data | Timeline of status, location, user, time; time between stages; test round times. |
| Actions | Scan or search; open from worklist and results. |
| Business rules | — |
| Depends on | Samples, Results |

## 4.5 Laboratory (5)

**CP-18** **Worklist** Laboratory R1

Lists the tests to perform per unit, with turnaround status.

| | |
|---|---|
| Actors | Technician, unit head |
| Key data | One row per test: patient no., accession, sample ID, patient, age/sex, test, test status, sample status, collection, expected result, delay, consumed %; filters by collection date, group, test, analyser, sample status, payer; status boxes; counters requests/tests/samples. |
| Actions | Search; print worklist; track sample; open results. |
| Business rules | Filtered by collection date; expected date = collection + duration counting test days; row colour by share of duration consumed (BR-13). |
| Depends on | Tests (timing), Samples, Parameters (thresholds) |

**CP-19** **Results by accession** Laboratory R1 filters auto-collapse, report units as tabs, no analyser column

Enters, reviews, verifies and prints results of a visit.

| | |
|---|---|
| Actors | Technician, unit head, microbiologist, pathologist |
| Key data | Query: accession, patient no., name (%), test status, sample status, from analyser, not printed / not verified / delta. Case: patient data, CR, preparations, diagnosis, flags. One tab per report unit (clinical group) showing its sample IDs and status. Grid: test, result, unit, reference range, flag, sample status, test status, delta, printed, attachments, logs. Report comment, internal notes. |
| Actions | Run query (filters collapse once a visit is open); Save; Review; Verify; Undo; Preview; Print; insert comment from library; culture entry; text entry with templates; previous results; attachments; result log; status log. |
| Business rules | Received gate (BR-03); ranges resolved by sex, age, condition (BR-08); panic values alert; restricted lists accept listed values only; formulas compute automatically; delta flag must be acknowledged before verify (BR-11); verify needs a value and the result right; preview refused while values are missing; only verified prints (BR-09); printed flag and status log on print; edits of verified results need the special right. |
| Depends on | Tests, Roles (result rights), Comment library, Report templates, Parameters |

**CP-20** **Results by sample** Laboratory R2

Works only the tests on a scanned tube.

| | |
|---|---|
| Actors | Technician |
| Key data | Sample ID → its report units (tabs). |
| Actions | Scan; same actions as results by accession. |
| Business rules | Same rules as CP-19. |
| Depends on | Results by accession, Samples |

**CP-21** **Work queue** Laboratory R2

Status-driven list of report units for the day's work and printing.

| | |
|---|---|
| Actors | Technician, unit head, reception |
| Key data | Registration-date range, group, payer, from/to analyser, normal/abnormal, received-only, keep-previous; status boxes exact (Complete, Reviewed, Verified, Printed) and cumulative Not-X; nine colour states; count; print list. |
| Actions | Search; open report (R); review all; verify all; add to print list; print list; abbreviated report; analyser closing presets. |
| Business rules | Not Verified = Ordered + Complete + Reviewed, Not Reviewed = Ordered + Complete, Not Complete = Ordered (BR-10); bulk actions apply to every listed result and are audited; print list numbers. |
| Depends on | Results, Analysers, Roles |

**CP-22** **Quality control** Laboratory R2

Placeholder for QC materials, targets, Levey-Jennings and Westgard rules.

| | |
|---|---|
| Actors | Technician, unit head |
| Key data | Analyser, test, level, value, target, SD index, rule, status. |
| Actions | View runs. |
| Business rules | Scope to be confirmed. |
| Depends on | Analysers, Tests |

## 4.6 Finance (13)

**CP-23** **Payments** Finance R1

Collects money on visits and prints receipts.

| | |
|---|---|
| Actors | Cashier |
| Key data | Visit search; total, discount, patient share, paid, refunded, remaining; payment: amount, method (the laboratory payment methods), reference; receipts. |
| Actions | Pay (full or partial); refund when the balance is negative; receipts and invoice; reprint. |
| Business rules | Amount within the remaining balance; reference mandatory when the payment method requires it; every payment stamped with cashier, branch, method and time; receipt auto-printed per parameter. |
| Depends on | Payment methods, Parameters |

### CP-24 Refunds Finance

R2 refund action also available on Payments in R1

Returns money when a visit balance is negative.

| | |
|---|---|
| Actors | Cashier with the Refund right |
| Key data | Negative balances; refund amount, method, reason; refund receipt. |
| Actions | Look up visit; refund; print receipt for signature. |
| Business rules | Refund never exceeds the refundable amount (BR-14); reason mandatory; audited. |
| Depends on | Payments, Reasons |

### CP-25 Money transfer Finance

R2

Moves a visit to another contract or shifts a line between patient and payer.

| | |
|---|---|
| Actors | Accounts, cashier with right |
| Key data | Visit, current and new payer/tier, re-priced lines, resulting balance; per-line actual amounts. |
| Actions | Contract transfer with preview; line transfer; apply. |
| Business rules | Re-pricing follows the new contract (BR-06); an overpayment becomes a refund; every transfer is logged. |
| Depends on | Payers, Contract pricing, Refunds |

### CP-26 Cancellations Finance

R2 available as an action on Search registrations in R1

Cancels services of a visit with a reason.

| | |
|---|---|
| Actors | Reception, accounts with right |
| Key data | Visit lines with status and shares; cancellation reason; cancelled list. |
| Actions | Select services; cancel. |
| Business rules | Verified services cannot be cancelled; related samples cancelled; receivables adjusted; nothing physically deleted (BR-15). |
| Depends on | Reasons, Samples |

**CP-27** **Batch payment** Finance R2

Settles a batch or lab-to-lab batch.

| | |
|---|---|
| Actors | Cashier |
| Key data | Batch no., members with shares, paid and remaining. |
| Actions | Pay member; pay all. |
| Business rules | Lab-to-lab batches are billed to the sending lab through claims. |
| Depends on | Batch registration, Lab-to-Lab registration, Payments |

**CP-28** **Treasuries (withdrawn)** Finance Withdrawn

2026-10-20, LIS-CR-004: the treasury concept was removed from the system

Withdrawn on 2026-10-20: the system no longer has cash drawers. Payments carry cashier and branch.

| | |
|---|---|
| Actors | — |
| Key data | — |
| Actions | — |
| Business rules | Existing treasury rows are kept in the database as history; nothing reads or writes them. |
| Depends on | |

**CP-29** **Treasury requests (withdrawn)** Finance Withdrawn

2026-10-20, LIS-CR-004: the treasury concept was removed from the system

Withdrawn on 2026-10-20 with CP-28: payments need no device approval.

| | |
|---|---|
| Actors | — |
| Key data | — |
| Actions | — |
| Business rules | Existing requests are kept in the database as history. |
| Depends on | |

### CP-30 Cash closing Finance R2

Reconciles the cash of a cashier (or every cashier of the branch) for a period.

| | |
|---|---|
| Actors | Cashier, manager |
| Key data | Payments and refunds list; totals: payments, cash, card/POS/transfer, refunds, expected cash, outstanding balances, discounts. |
| Actions | Filter; print closing sheet. |
| Business rules | Expected cash = cash payments − refunds; card and transfer amounts are not in the drawer (BR-16). |
| Depends on | Payments, Refunds |

### CP-31 Claims Finance R2

Bills credit payers for a period.

| | |
|---|---|
| Actors | Accounts |
| Key data | Payer, tier (or all), period, layout (standard, per tier, with results, totals only, with counts, credit note); claim no., visits, total, status Draft/Issued. |
| Actions | Preview; save draft; issue; print; Excel. |
| Business rules | Cash payers never receive claims; claim totals are the payer shares of the visits in the period. |
| Depends on | Payers, Contract pricing, Visits |

### CP-32 Payers & contracts Finance R1

Payers (ranks) and their contract tiers.

| | |
|---|---|
| Actors | Accounts |
| Key data | Name (EN/AR), search key, type (cash, credit, insurance, lab), web-results portal credentials, tiers (EN/AR). |
| Actions | Add, edit; open contract pricing. |
| Business rules | Web username mandatory when the portal is enabled; payer with visits cannot be deleted. |
| Depends on | Contract pricing |

### CP-33 Base price lists

Finance

R1

Named lists holding the price per service.

| | |
|---|---|
| Actors | Accounts |
| Key data | Code, name, year, active flag; price and active flag per service; derived contracts. |
| Actions | New list; edit prices inline; copy list with × factor or + amount; Apply. |
| Business rules | Contract prices refresh only on Apply (BR-07); inactive services do not appear at ordering. |
| Depends on | Services, Contract pricing |

### CP-34 Contract pricing

Finance

R2 Release 1 uses the seeded contract formulas

Derives each tier's prices from base lists by formula.

| | |
|---|---|
| Actors | Accounts |
| Key data | Payer, tier, effective dates, up to three base lists with patient and payer factors, apply date; derived price grid with exclusion and alert condition per service. |
| Actions | Edit formula; Apply; exclude a negotiated price; set alert; copy from another tier; review report. |
| Business rules | Patient share + payer share per service (BR-04); excluded prices survive Apply; alerts pop up at ordering. |
| Depends on | Base price lists, Payers |

### CP-35 Financial reports

Finance

R2

Sixteen operational finance reports.

| | |
|---|---|
| Actors | Accounts, management |
| Key data | Patients' income, payment methods, contract type, monthly statistics, patient statement, doctor income with share, clients income, service income analysis, outstanding balances, discounts, cancellations, transfers, total per day, price-list hand-out, cost vs income, turnaround compliance. |
| Actions | Run with period and branch; print; Excel. |
| Business rules | Figures reconcile with payments, refunds and visit shares. |
| Depends on | Visits, Payments, Doctors, Tests |

## 4.7 Setup (22)

**CP-36** **Branches** Setup R1

Laboratory branches.

| | |
|---|---|
| Actors | Lab administrator |
| Key data | Code, EN/AR names, country, governorate, phone, email, currency, status. |
| Actions | Add, edit, deactivate. |
| Business rules | Branch with visits cannot be deleted. |
| Depends on | Countries & governorates |

**CP-37** **Roles & privileges** Setup R1

Privilege groups users inherit.

| | |
|---|---|
| Actors | Lab administrator |
| Key data | Name (EN/AR), description, status; per screen Query/Insert/Update/Delete; special rights (discount with cap, payment, refund, transfer, cancel, modify verified result, preview without verify, print invoice, VIP…); result rights See/Review/Verify/Edit/Print per clinical group and branch; locations. |
| Actions | Add, edit, grant/revoke all, filter screens. |
| Business rules | Active role needs a screen; role with users cannot be deleted; changing a role changes all its users (BR-17). |
| Depends on | Clinical groups, Branches |

**CP-38** **Users** Setup R1

Staff accounts.

| | |
|---|---|
| Actors | Lab administrator |
| Key data | Name (EN/AR), username, mobile, email, designation, role, branches, shifts, status; password policy, expiry, must-change, lock; signature image; privilege overrides with restore. |
| Actions | Add, edit, lock, restore to role. |
| Business rules | Username 4–30 chars; overrides affect one user only; main administrator protected. |
| Depends on | Roles, Branches |

**CP-39** **Clinical groups** Setup R1

Laboratory units that split tubes, rights and reports.

| | |
|---|---|
| Actors | Lab administrator |
| Key data | Code, unit code, EN/AR names, report name, report template, display order, status. |
| Actions | Add, edit, deactivate. |
| Business rules | Group with services cannot be deleted. |
| Depends on | Report templates |

**CP-40** **Services** Setup R1

Orderable items: tests/profiles, packages, cultures, text reports, additional services.

| | |
|---|---|
| Actors | Lab administrator |
| Key data | Code, short name, shortcut no., EN/AR names, report name, LOINC, CPT, group, type, sample type and condition, header, aliases, precaution, gender, duration, labels, print-alone, visibility flags, status; members with order, in-sum and visibility; profile sum; base price. |
| Actions | Add, edit, manage members. |
| Business rules | At least one member; unique member order; package members are services; print-alone services print on their own sheet. |
| Depends on | Tests, Sample types, Clinical groups, Base price lists |

### CP-41 Tests Setup R1

Measured parameters and their configuration.

| | |
|---|---|
| Actors | Lab administrator, unit head |
| Key data | Code, EN/AR names, report name, LOINC, CPT, group, type (numeric, non-numeric, optional list, restricted list, formula, formatted text, culture), sample type and condition, unit, gender, methodology, decimals, flags; parent services (the services that carry the test, with its order in each); numeric range bands by sex/age/condition with normal and panic limits, reportable limits; list values with normal/critical/default/colour; formula; text paragraphs; duration, minimum duration, frequency check, delta %, test days; labels, prefix, print reference. |
| Actions | Add, edit, configure per type, check formula; add the test to a profile or culture, or remove it, from the parent-services block. |
| Business rules | Default list value must not read as a result (BR-19); formulas validated against test codes; hidden helper tests never print (BR-20); parent services follow the member rules of each service type: a test service or text report keeps its single test and a package takes none (set up from the service), profiles and cultures are joined or left from the test. |
| Depends on | Clinical groups · Sample types · Units · Patient conditions · Services |

### CP-42 Service sequence Setup R1

Global display and report order of services.

| | |
|---|---|
| Actors | Lab administrator |
| Key data | Sequence per service. |
| Actions | Edit inline. |
| Business rules | Unique and greater than zero. |
| Depends on | Services |

### CP-43 Sample types Setup R1

Sample types and their conditions.

| | |
|---|---|
| Actors | Lab administrator |
| Key data | Code, short name, EN/AR names, conditions (fasting, random, refrigerated…), status. |
| Actions | Add, edit. |
| Business rules | Type referenced by tests cannot be deleted. |
| Depends on | — |

**CP-44** **Units of measurement** Setup R1

Result units.

| | |
|---|---|
| Actors | Lab administrator |
| Key data | Name, status. |
| Actions | Add, edit. |
| Business rules | Unit referenced by tests cannot be deleted. |
| Depends on | — |

**CP-45** **Patient conditions** Setup R2 seeded in Release 1

Conditions that switch reference ranges.

| | |
|---|---|
| Actors | Lab administrator |
| Key data | EN/AR names, status. |
| Actions | Add, edit. |
| Business rules | Used by registration and range bands (BR-08). |
| Depends on | Tests |

**CP-46** **Preparations** Setup R1

Questions asked at registration.

| | |
|---|---|
| Actors | Lab administrator |
| Key data | Code, EN/AR question, type (text, yes/no, option list), options, gender, linked services, status. |
| Actions | Add, edit. |
| Business rules | Option list needs at least two options; empty service link = all services. |
| Depends on | Services |

### CP-47 Comment library Setup R1

Reusable report comments.

| | |
|---|---|
| Actors | Unit head |
| Key data | Short name, EN/AR text, group scope, status. |
| Actions | Add, edit. |
| Business rules | Inserted into report comments with % search. |
| Depends on | Clinical groups |

### CP-48 Doctors Setup R1

Referring doctors.

| | |
|---|---|
| Actors | Lab administrator |
| Key data | Code, EN/AR name, specialty, occupation, branches, share %, phone, status. |
| Actions | Add, edit. |
| Business rules | Share % feeds the doctor-income report. |
| Depends on | Branches |

### CP-49 Diagnoses Setup R2 seeded in Release 1

ICD codes with linked services.

| | |
|---|---|
| Actors | Lab administrator |
| Key data | ICD code, EN/AR name, services, status. |
| Actions | Add, edit. |
| Business rules | Choosing a diagnosis at registration adds its services. |
| Depends on | Services |

**CP-50** **Sending labs** Setup R2

Contracted laboratories sending samples.

| | |
|---|---|
| Actors | Lab administrator |
| Key data | EN/AR name, payer, tier, courier (EN/AR), phone, doctor, status. |
| Actions | Add, edit. |
| Business rules | — |
| Depends on | Payers |

**CP-51** **Shortcut sets** Setup R2 seeded in Release 1

Typing shortcuts dropping several services.

| | |
|---|---|
| Actors | Lab administrator |
| Key data | Code, EN/AR name, services, status. |
| Actions | Add, edit. |
| Business rules | Each service keeps its own price; lines remain editable. |
| Depends on | Services |

**CP-52** **Test rules** Setup R2 seeded in Release 1

Auto-add rules.

| | |
|---|---|
| Actors | Lab administrator |
| Key data | EN/AR name, group, trigger services, added services, status. |
| Actions | Add, edit, validate. |
| Business rules | A service cannot be both trigger and added; added services are charged at their own price. |
| Depends on | Services |

### CP-53 Reasons Setup

R2 seeded in Release 1

Reason lists.

| | |
|---|---|
| Actors | Lab administrator |
| Key data | Kind (rejection, cancellation, discount, refund), EN/AR name, status. |
| Actions | Add, edit. |
| Business rules | — |
| Depends on | — |

### CP-54 Report templates Setup

R2 seeded in Release 1

Print templates.

| | |
|---|---|
| Actors | Lab administrator |
| Key data | Name, type (results, culture, text, claim, receipt, label, job order), logo, signature, QR, language, header text, status. |
| Actions | Add, edit. |
| Business rules | — |
| Depends on | — |

### CP-55 Additional services Setup

R2 seeded in Release 1

Chargeable non-test services.

| | |
|---|---|
| Actors | Lab administrator |
| Key data | Code, short name, EN/AR name, shortcut no., status, base price. |
| Actions | Add, edit. |
| Business rules | No sample and no result; not on the job order. |
| Depends on | Base price lists |

**CP-56** **Drug types · Antibiotics · Organisms** Setup R1

Microbiology masters for culture entry.

| | |
|---|---|
| Actors | Microbiologist, lab administrator |
| Key data | Drug types; antibiotics (code, name, abbreviation, commercial name); organisms (code, name). |
| Actions | Add, edit. |
| Business rules | — |
| Depends on | — |

**CP-57** **Parameters** Setup R2 Release 1 runs on the defaults

Tenant behaviour switches.

| | |
|---|---|
| Actors | Lab administrator |
| Key data | Mandatory mobile / doctor / national ID; similar-name warning; discount cap; accession format; auto-print labels / job order / receipt; print unverified; label copies; result language; auto-collect; auto-receive; new barcode on reject; review before verify; delay thresholds; session timeout. |
| Actions | Toggle; edit. |
| Business rules | Read by every module at runtime. |
| Depends on | — |

## 4.8 Integration (5)

**CP-58** **Analysers** Integration R2

Connected instruments.

| | |
|---|---|
| Actors | Integration engineer |
| Key data | Code, name, group, default sample, mode (uni, bi-directional, download, shared folder), serial or TCP settings, auto-print, download orders, append unregistered tests/samples, overwrite reviewed/verified, print copies, flags (stop / record), status, connection. |
| Actions | Add, edit, test connection, open mappings. |
| Business rules | Serial settings must look like 9600,n,8,1; flags marked "stop" hold results in auto-verification. |
| Depends on | Clinical groups, Sample types, Site agent |

**CP-59** **Analyser tests** Integration R2

Maps instrument codes to tests.

| | |
|---|---|
| Actors | Integration engineer |
| Key data | Per mapping: test, analyser code, host code, result selection, slope, intercept, unit, query, query details, Auto Send, conversion table, analyser-specific range, methodology. |
| Actions | Map test; edit inline; conversion table; range override; print code sheet. |
| Business rules | Auto Send is the permission to save incoming results; a mapping without it never lands a result. |
| Depends on | Analysers, Tests |

**CP-60** **Analyser samples** Integration R2

Sample-type codes per instrument.

| | |
|---|---|
| Actors | Integration engineer |
| Key data | Analyser, sample type, code. |
| Actions | Add, edit. |
| Business rules | — |
| Depends on | Analysers, Sample types |

**CP-61** **Site agent** Integration R2

Per-branch connector status and logs.

| | |
|---|---|
| Actors | Integration engineer, support |
| Key data | Agent name, branch, version, status, heartbeat, messages and errors, connected analysers, configuration, decrypted hourly log. |
| Actions | Push configuration; fetch logs. |
| Business rules | Happy path shows StartSend then Passed Check per sample ID; an Exception line names the cause. |
| Depends on | Analysers, Branches |

**CP-62** **PCR kits** Integration R2

Interpretation per instrument and kit.

| | |
|---|---|
| Actors | Integration engineer |
| Key data | Kit, instrument, serial lock, run-file layout, value column, per-gene positive/borderline thresholds, verdict rules, result/backup/failed folders, status. |
| Actions | Add, edit, view, simulate run. |
| Business rules | A low Ct is positive; all genes negative including the internal control = INVALID; files from other instruments are refused. |
| Depends on | Analysers |

## 4.9 Automation (2)

**CP-63** **Verification settings** Automation R2

Auto-verification engine switches.

| | |
|---|---|
| Actors | Unit head, lab administrator |
| Key data | Master switch; per clinical group the checkers (medical rules, no-rules guard, result, comment, option, flag, device flag, incomplete, panic, normal, delta, all-group-complete, inter-analyser delta, reportable, QC); result options; result dictionary. |
| Actions | Enable/disable; toggle checkers; simulate rules. |
| Business rules | Normal-range checker overrides rules and stays off by default; enable only after simulation, group by group. |
| Depends on | Medical rules, Tests, Analysers |

**CP-64** **Medical rules** Automation R2

Release rules per test.

| | |
|---|---|
| Actors | Unit head |
| Key data | Group, test, parameters; rules with formula condition, formula, recommendation, active; operator keypad. |
| Actions | New rule; edit; check rule; print rules. |
| Business rules | Percentages written as multiplications of the live range (BR-21); every boundary needs its complement (BR-22); last-result rules need No LR (BR-23); a parameter must carry a LOINC code to be usable. |
| Depends on | Tests, Verification settings |

## 4.10 Account (4)

**CP-65** **Profile** Account R1

Laboratory and credential details.

| | |
|---|---|
| Actors | Lab administrator |
| Key data | Lab name, portal URL, country, plan, manager, mobile, email; username, password. |
| Actions | Save; update credentials via OTP. |
| Business rules | — |
| Depends on | Subscriptions (SA) |

**CP-66** **Wallet** Account R2

Prepaid balance and usage.

| | |
|---|---|
| Actors | Lab administrator |
| Key data | Balance, next charge, SMS used, transactions. |
| Actions | Request top-up. |
| Business rules | Metered services pause at zero; the portal keeps working. |
| Depends on | Subscriptions (SA) |

**CP-67** **Support** Account R2

Support tickets.

| | |
|---|---|
| Actors | Lab administrator |
| Key data | Ticket no., subject, priority, status, created. |
| Actions | New ticket. |
| Business rules | — |
| Depends on | — |

**CP-68** **Audit log** Account R2

Tenant audit trail.

| | |
|---|---|
| Actors | Lab administrator |
| Key data | Time, user, action, entity, details, branch. |
| Actions | Filter; export. |
| Business rules | Immutable. |
| Depends on | All pages |

# 5. Super Admin Portal — pages

## 5.1 Access (1)

**SA-01** **Login** Access R1

Authenticates National Technology staff at the Super Admin address <base>/admin/login.

| | |
|---|---|
| Actors | Super Admin staff |
| Key data | Username, password. |
| Actions | Login; Forgot password. |
| Business rules | Only platform staff sign in there (BR-28); lock and expiry respected. |
| Depends on | Admin users |

## 5.2 Overview (2)

**SA-02** **Home** Overview R1

Landing page with the brand.

| | |
|---|---|
| Actors | Super Admin staff |
| Key data | — |
| Actions | — |
| Business rules | — |
| Depends on | — |

**SA-03** **Dashboard** Overview R1

Platform KPIs.

| | |
|---|---|
| Actors | Super Admin staff |
| Key data | Clients active/inactive, admin users, served patients, tests and registrations per period, wallets at zero, subscription vs manual, plan distribution, registrations by client, recent clients. |
| Actions | Change period; refresh. |
| Business rules | — |
| Depends on | Clients Subscriptions |

## 5.3 Administration (6)

**SA-04** **Clients** Administration R1

Tenant lifecycle.

| | |
|---|---|
| Actors | Operations Admin |
| Key data | Name (EN/AR), country, governorate, address, mobile, licence, manager name and mobile, email, plan, portal URL, wallet, branches and users counts, created, source (subscription / manual), status. |
| Actions | Add; edit; reset password; open wallet; activate / deactivate; filters by source and plan. |
| Business rules | Creating a client copies the global master data (BR-25); deactivation blocks all client users at login. |
| Depends on | Countries, Governorates, Plans, Master data |

**SA-05** **Subscriptions & wallets** Administration R1

Plans, wallet balances and transactions.

| | |
|---|---|
| Actors | Operations Admin |
| Key data | Plans (fee, branches, users, SMS quota, analysers); client wallet balance, next charge, transactions (top-up, subscription, usage). |
| Actions | Top up; select client. |
| Business rules | Monthly fee and metered usage debit the wallet; zero balance pauses metered services. |
| Depends on | Clients |

**SA-06** **Roles** Administration R1

Platform roles.

| | |
|---|---|
| Actors | Super Admin |
| Key data | Name, description, status; per-screen Query/Insert/Update/Delete; platform special rights (impersonate client admin, reset client password, activate client, expire passwords). |
| Actions | Add, edit. |
| Business rules | Active role needs a screen; role with users protected; no per-user override. |
| Depends on | — |

**SA-07** **Admin users** Administration R1

National Technology staff accounts.

| | |
|---|---|
| Actors | Super Admin |
| Key data | Full name, username, mobile, email, role, password expiry, lock, status. |
| Actions | Add, edit, lock. |
| Business rules | Username 4–30 chars; password policy; privileges inherited from the role only; main administrator protected. |
| Depends on | Roles |

**SA-08** **Sessions** Administration R1

Active admin sessions.

| | |
|---|---|
| Actors | Super Admin |
| Key data | User, IP, since, status. |
| Actions | Terminate. |
| Business rules | — |
| Depends on | Admin users |

**SA-09** **Audit log** Administration R1

Platform audit trail.

| | |
|---|---|
| Actors | Super Admin |
| Key data | Time, user, action, entity, details, client. |
| Actions | Filter; export. |
| Business rules | Immutable. |
| Depends on | All pages |

## 5.4 Master data (4)

**SA-10** **Countries** Master data R1

Countries and their validation data.

| | |
|---|---|
| Actors | Operations Admin |
| Key data | Name (EN/AR), code, dial code, currency and code, language, time zone, mobile length, flag, remarks, status. |
| Actions | Add, edit. |
| Business rules | Dial code like +20; country referenced by governorates cannot be deleted. |
| Depends on | — |

**SA-11** **Governorates** Master data R1

Governorates and districts.

| | |
|---|---|
| Actors | Operations Admin |
| Key data | Name (EN/AR), country, district, remarks, status. |
| Actions | Add, edit. |
| Business rules | — |
| Depends on | Countries |

**SA-12** **Clinical groups · Services · Tests · Service sequence · Sample types · Units · Preparations · Comment library**

Master data R1

Global catalogue templates with the same editors as the client portal.

| | |
|---|---|
| Actors | Master Data Reviewer, Operations Admin |
| Key data | As CP-39 to CP-47. |
| Actions | Add, edit, publish. |
| Business rules | Changes reach tenants according to the master-data update policy (SA-33). |
| Depends on | Platform parameters |

**SA-13** **Patient conditions · Reasons · Diagnoses · Additional services · Report templates · Test rules · Shortcut sets**

Master data R1

Global versions of the catalogue-depth lists.

| | |
|---|---|
| Actors | Operations Admin |
| Key data | As CP-45, CP-49, CP-51 to CP-55. |
| Actions | Add, edit, publish. |
| Business rules | Same publication policy. |
| Depends on | Services Tests |

## 5.5 Microbiology (1)

**SA-14** **Drug types · Commercial names · Antibiotics · Organisms · Sensitivity values** Microbiology R1

Global microbiology masters.

| | |
|---|---|
| Actors | Master Data Reviewer |
| Key data | Drug types; commercial names with drug type; antibiotics (code, name, abbreviation, commercial name); organisms (code, name); sensitivity values (S/I/R). |
| Actions | Add, edit. |
| Business rules | — |
| Depends on | — |

## 5.6 Integration library (4)

**SA-15** **Analyser models** Integration library R2

Vendor/model library tenants choose from.

| | |
|---|---|
| Actors | Integration team |
| Key data | Vendor, model, discipline, protocol, physical link, modes, driver version, status. |
| Actions | Add, edit. |
| Business rules | — |
| Depends on | — |

**SA-16** **PCR kits** Integration library R2

Kit interpretation library.

| | |
|---|---|
| Actors | Integration team |
| Key data | As CP-62. |
| Actions | Add, edit, view. |
| Business rules | — |
| Depends on | Analyser models |

**SA-17** **Code systems** Integration library R2

LOINC, CPT, ICD-10, SNOMED versions.

| | |
|---|---|
| Actors | Integration team |
| Key data | Name, version, items, use, status. |
| Actions | Import codes. |
| Business rules | — |
| Depends on | — |

**SA-18** **Rule templates** Integration library R2

Reusable medical-rule patterns.

| | |
|---|---|
| Actors | Integration team |
| Key data | Name, group, formula, description, status. |
| Actions | Add, edit. |
| Business rules | — |
| Depends on | — |

## 5.7 Platform (3)

**SA-19** **Platform parameters** Platform R1

Platform-wide defaults.

| | |
|---|---|
| Actors | Super Admin |
| Key data | OTP length and validity, password minimum, session timeout, trial days, default currency and language, master-data update policy, default delay thresholds and discount cap. |
| Actions | Save. |
| Business rules | The master-data policy decides whether later global changes are pushed, proposed or ignored for tenants. |
| Depends on | — |

**SA-20** **Notification templates** Platform R1

SMS and email texts.

| | |
|---|---|
| Actors | Super Admin |
| Key data | Code, channel, EN/AR text with placeholders, status. |
| Actions | Add, edit. |
| Business rules | Placeholders: {code} {url} {patient} {acc} {lab} {test} {value}. |
| Depends on | — |

**SA-21** **Profile** Platform R1

Administrator details and credentials.

| | |
|---|---|
| Actors | Super Admin staff |
| Key data | Name, username, mobile, email; password. |
| Actions | Save; update credentials via OTP. |
| Business rules | — |
| Depends on | Admin users |

# 6. Status models

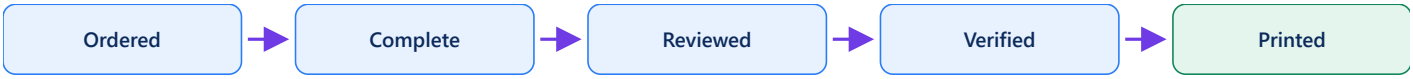
### Sample (tube)




Rejected (with reason) → Unrejected or Replaced with a new barcode

Collected on scan at phlebotomy, Received on scan at the unit; auto-collect / auto-receive parameters set both on save. Results need Received (BR-03).

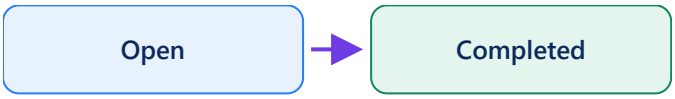
### Test / result




Undo returns Verified/Reviewed to the previous state (audited)

A report unit takes the lowest status of its results; a profile advances only when every member has (BR-04). Only Verified prints (BR-09).

### Visit




Services may be Reserved (deferred) or Cancelled (with reason)

Completed when every report unit is printed and the balance is settled.

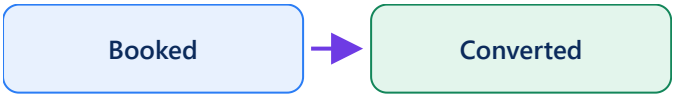
### Payment balance




Negative balance → Refund due

Remaining = patient share − payments + refunds (BR-14).

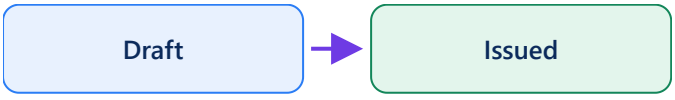
### Home booking




Cancelled

Conversion creates the visit.

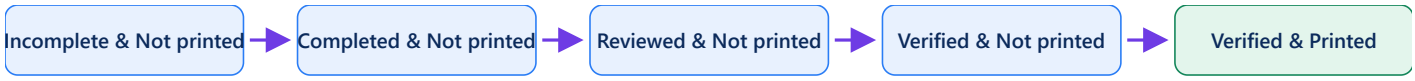
### Claim




—

Issued claims are printed and sent to the payer.

### Work-queue colour states




Incomplete & Printed · Completed & Printed · Reviewed & Printed · Reserved & Not printed

Nine colours per installation; the meaning is fixed.

# 7. End-to-end workflows

## 7.1 WF-01 — Laboratory onboarding

Pages: CP-02 SA-04 CP-36 CP-37 CP-38 CP-33 CP-32 Release: R1

| # | Actor | Action | System response |
|---|---|---|---|
| 1 | **Laboratory manager** | Completes the subscription wizard (lab details, OTP, credentials). | Creates the tenant, copies the global master data, generates the portal URL and sends it by SMS. |
| 2 | **Operations Admin (alternative)** | Creates the client manually and activates it. | Same provisioning; source marked Manual. |
| 3 | **Lab administrator** | Logs in, selects the branch, creates branches, roles and users. | Users inherit role privileges. |
| 4 | **Lab administrator / accounts** | Reviews the inherited catalogue, sets base prices and payers with tiers. | Contract prices derive by formula on Apply. |
| 5 | **Reception** | Registers the first patient. | Labels, job order and receipt print; the laboratory is live. |

## 7.2 WF-02 — Walk-in patient — registration to receipt

Pages: CP-06 CP-10 CP-23 Release: R1

| # | Actor | Action | System response |
|---|---|---|---|
| 1 | **Reception clerk** | Types the patient name. | Shows similar existing patients with number and mobile (BR-01). |
| 2 | **Reception clerk** | Chooses the existing file or confirms a new patient; completes title, gender, date of birth or age, mobile, condition. | Title fills gender; age and birth date sync; patient number assigned on save. |
| 3 | **Reception clerk** | Sets payer and contract tier, doctor, flags, diagnosis. | Diagnosis adds its service set; prices follow the contract (BR-06). |
| 4 | **Reception clerk** | Adds services by name, number, set or picker. | Refuses duplicates and gender-restricted services; alerts or blocks repeats (BR-12); shows precautions and contract alerts; adds companion services by rule; shows patient and payer shares and expected result dates. |

| # | Actor | Action | System response |
|---|---|---|---|
| 5 | **Reception clerk** | Sets a service to Reserved if the precaution is unmet; applies a discount with reason; answers preparation questions. | Reserved lines stay billed without a sample (BR-05); discount limited by the user cap. |
| 6 | **Reception clerk (R1)** | Chooses the payment method and amount. | Defaults to the amount due; partial amounts allowed; reference required for non-cash. |
| 7 | **Reception clerk** | Saves (Ctrl+S). | Creates the accession, generates tubes by group + sample type + condition (BR-02), creates result rows, records the payment with cashier and branch, prints labels, job order and receipt per parameters; shows the success dialog. |
| 8 | **Reception clerk (R2)** | Collects the payment from the success dialog or the Payments page. | Receipt printed; balance updated. |

## 7.3 WF-03 — Home visit

Pages: CP-07 CP-06 Release: R2

| # | Actor | Action | System response |
|---|---|---|---|
| 1 | **Reception clerk** | Books the visit with address, time, phlebotomist and services. | Booking number; preliminary labels and job order; quote. |
| 2 | **Phlebotomist** | Visits the patient and collects the tubes. | — |
| 3 | **Reception clerk** | Converts the booking. | Registration pre-filled; collector set to the phlebotomist; booking marked Converted. |

## 7.4 WF-04 — Batch and lab-to-lab registration

Pages: CP-08 CP-09 CP-27 CP-31 Release: R2

| # | Actor | Action | System response |
|---|---|---|---|
| 1 | **Reception clerk** | Enters the group contract and member rows, or chooses the sending lab and its rows. | One accession per row under a batch number; collected-outside flag on lab-to-lab rows. |
| 2 | **Reception clerk** | Saves the batch. | Labels and job orders for all members. |
| 3 | **Cashier / accounts** | Settles the batch or lets claims bill the sending lab. | Pay-all distributes across members; lab batches appear on claims. |

## 7.5 WF-05 — Sample lifecycle

Pages: CP-14 CP-15 CP-16 CP-17 Release: R2

| # | Actor | Action | System response |
|---|---|---|---|
| 1 | **Phlebotomist** | Scans the tube after drawing. | Status Collected with time, user and location. |
| 2 | **Technician** | Scans the tube at the unit. | Status Received; results may now be entered (BR-03). |
| 3 | **Technician** | Rejects an unfit tube with a reason. | Status Rejected; reception notified to recollect; tests take no results. |
| 4 | **Technician** | Reinstates or replaces the tube. | Same barcode back to Received, or a new sample with a new barcode (BR-18). |
| 5 | **Anyone** | Opens sample tracking. | Timeline, time between stages, test round times. |

## 7.6 WF-06 — Result entry, review, verification and printing

Pages: CP-18 CP-19 CP-20 CP-21 Release: R1 (R2 for sample view and queue)

| # | Actor | Action | System response |
|---|---|---|---|
| 1 | **Technician** | Opens the worklist for the unit. | One row per test with expected result date and delay colour (BR-13). |
| 2 | **Technician** | Opens the visit on Results by accession (or scans the tube). | Filters collapse; one tab per report unit with its sample IDs and status. |
| 3 | **Technician** | Enters values, picks list values, leaves formulas to compute. | Ranges resolved by sex, age and condition (BR-08); flags and panic alerts; Restricted lists validated; delta flag against the previous result; status Complete on save. |
| 4 | **Technician** | Adds a report comment from the library; writes an internal note if needed. | Report comment prints; notes never print. |
| 5 | **Technician / unit head** | Reviews, then verifies. | Rights per clinical group checked; verify refused without a value; delta flag must be acknowledged (BR-11); electronic signature recorded. |
| 6 | **Reception / technician** | Previews and prints. | Preview refused while values are missing; only verified prints (BR-09); printed flag and status log updated; print list number assigned in bulk printing. |

## 7.7 WF-07 — Analyser and PCR results

Pages: CP-58 CP-59 CP-61 CP-62 CP-21 Release: R2

| # | Actor | Action | System response |
|---|---|---|---|
| 1 | **Integration engineer** | Defines the analyser, maps its codes with Auto Send, configures the site agent. | Agent connects over serial, TCP or file share. |
| 2 | **Analyser** | Sends results for a scanned sample ID. | Results land as Complete with the analyser source; flags recorded; conversion tables applied; unmapped or Auto-Send-off codes refused and logged. |
| 3 | **PCR instrument** | Exports a run file to the result folder. | Kit interpretation produces DETECTED / NOT DETECTED / Presumptive / INVALID per sample; file moved to backup or failed. |
| 4 | **Chemist** | Closes the analyser's work from the queue (from-analyser, then to-analyser). | Everything produced is reviewed; everything pending is traced. |

## 7.8 WF-08 — Auto-verification rollout

Pages: CP-63 CP-64 Release: R2

| # | Actor | Action | System response |
|---|---|---|---|
| 1 | **Unit head** | Gives the release rules per group in medical language. | — |
| 2 | **Lab administrator** | Writes the rules with the keypad, using formula conditions and No LR where needed; checks each rule. | Rule validated (BR-21 to BR-23). |
| 3 | **Lab administrator** | Simulates the rules on recent results. | Released / held counts with reasons. |
| 4 | **Lab administrator** | Enables the engine for one group at a time. | Incoming results pass the checkers; normal results release; the rest reach the doctor. |

## 7.9 WF-09 — Money movements and daily closing

Pages: CP-23 CP-24 CP-25 CP-26 CP-30 Release: R1 (payments, refunds) · R2 (rest)

| # | Actor | Action | System response |
|---|---|---|---|
| 1 | **Cashier** | Collects full or partial payments. | Receipt per payment; remaining balance tracked. |
| 2 | **Accounts** | Transfers a visit to a contract after approval arrives, or shifts a line value. | Re-pricing; excess becomes refundable. |
| 3 | **Cashier** | Refunds a negative balance with reason. | Refund receipt for patient signature; audited. |

| # | Actor | Action | System response |
|---|---|---|---|
| 4 | **Reception** | Cancels a service with reason. | Receivables adjusted; samples cancelled; record kept (BR-15). |
| 5 | **Cashier** | Closes the drawer. | Expected cash = cash payments − refunds (BR-16); handover sheet printed. |

## 7.10 WF-10 — Pricing maintenance and claims

Pages: CP-33 CP-32 CP-34 CP-31 CP-35 Release: **R1 (lists, payers) · R2 (pricing page, claims)**

| # | Actor | Action | System response |
|---|---|---|---|
| 1 | **Accounts** | Creates next year's base list by copying with a factor. | All prices derived in one step. |
| 2 | **Accounts** | Points each tier at the new list with patient and payer factors and applies. | Contract prices refresh; excluded prices kept (BR-07). |
| 3 | **Accounts** | Issues the monthly claim per credit payer. | Visits and payer shares listed in the chosen layout; draft then issued. |
| 4 | **Management** | Runs the financial reports. | Income, debtors, discounts, cancellations, turnaround. |

## 7.11 WF-11 — Privileges administration

Pages: CP-37 CP-38 Release: **R1**

| # | Actor | Action | System response |
|---|---|---|---|
| 1 | **Lab administrator** | Agrees the team split with the lab manager. | — |
| 2 | **Lab administrator** | Creates roles with screen rights, special rights and result rights per group and branch. | Users joined to a role inherit everything (BR-17). |
| 3 | **Lab administrator** | Adjusts one user if needed. | Marked modified; restore returns the user to the role. |

## 7.12 WF-12 — Platform operations

Pages: SA-04 SA-05 SA-12 SA-13 SA-19 Release: **R1**

| # | Actor | Action | System response |
|---|---|---|---|
| 1 | **Operations Admin** | Creates or activates clients, resets passwords, tops up wallets. | Audit entries; client users blocked when inactive. |

| # | Actor | Action | System response |
|---|---|---|---|
| 2 | **Master Data Reviewer** | Maintains the global catalogue and lists. | Changes published to tenants according to the policy. |
| 3 | **Super Admin** | Sets platform parameters and notification templates. | OTP, password and session rules applied platform-wide. |

# 8. Business rules

| ID | Rule | Statement |
|---|---|---|
| **BR-01** | Immutable identifiers | Patient numbers are generated by the system and never change; accessions are sequential per visit; sample IDs are unique and never reused. |
| **BR-02** | Tube merge | Services with the same clinical group, sample type and sample condition share one sample; any difference splits them. |
| **BR-03** | Received gate | A result may not be recorded until the sample status is Received (auto-receive emulates it in Release 1). |
| **BR-04** | Profile completion | A profile or package reaches Complete, Reviewed or Verified only when every member test has. |
| **BR-05** | Reserved keeps the bill | Deferring a service keeps it in the visit and the bill with no sample; cancelling removes it from both. |
| **BR-06** | Contract sets the price | Payer and contract tier determine every service's patient share and payer share. |
| **BR-07** | Three-layer pricing | Base list price × tier factors = contract price; Apply propagates base-list changes to all derived tiers except excluded services. |
| **BR-08** | Patient condition selects the range | Reference ranges resolve by sex, age and patient condition. |
| **BR-09** | Verify is a signature | Verification records who signed and when; only verified results print. |
| **BR-10** | Not-X queue logic | "Not X" returns everything below X in the test track. |
| **BR-11** | Delta check warns | A delta flag must be investigated and acknowledged but does not block verification. |
| **BR-12** | Frequency check | A repeat order inside the configured interval alerts or is prevented. |
| **BR-13** | Turnaround clock | Expected result = collection + duration counting only test days; delay colours are a percentage of the duration consumed. |
| **BR-14** | Negative balance means refund | A negative visit balance is returned through the refund flow, never above the refundable amount. |
| **BR-15** | Nothing is erased | Cancellations, transfers, rejections and edits keep the original data with reason, user and time. |

| ID | Rule | Statement |
|---|---|---|
| **BR-16** | Drawer equation | Expected cash = cash payments − refunds; card and transfer amounts are not in the drawer. |
| **BR-17** | Group privileges propagate | Changing a role changes every user in it; changing one user affects that user only. |
| **BR-18** | New sample, new barcode | A replacement sample carries a new barcode unless the tenant policy allows reinstatement. |
| **BR-19** | Default list value is not a result | The default of a list test is empty or a placeholder, never a value that reads as a result. |
| **BR-20** | Hidden helpers | Helper tests feeding formulas or sums never print. |
| **BR-21** | Percentage tolerance is a multiplication | A ±20 % tolerance is written as × 0.8 and × 1.2 of the live range values. |
| **BR-22** | Every boundary needs its complement | A rule for one side of a boundary needs a condition or complement so other patients are not stopped. |
| **BR-23** | No last result must not stop | Rules using the previous result must not stop patients who have none in the window. |
| **BR-24** | Validated selection only | Fields that reference master data accept only selected values. |
| **BR-25** | Tenant copy of master data | A tenant's master data is an independent copy of the global templates; later global changes follow the update policy. |
| **BR-26** | Payment inside registration (Release 1) | The amount captured on the registration page is recorded with the visit on save; it may not exceed the amount due; non-cash payments need a reference. |
| **BR-27** | Treasury per device (withdrawn) | Withdrawn on 2026-10-20 (LIS-CR-004): a money movement is stamped with the cashier, the session branch, the payment method and its reference; no cash drawer or device approval is required. |
| **BR-28** | Sign-in by address | Each laboratory signs in only at its own address (<base>/lab/<slug>) and the Super Admin only at <base>/admin/login; a user who is not a member of the laboratory of the address, or not platform staff at the admin address, is refused exactly like a wrong password. After sign-in the session carries the laboratory; signing out or timing out returns to the same address. |

# 9. Conceptual data model

| Area | Entities |
|---|---|
| **Platform** | Client (tenant) · Plan · Wallet transaction · Admin user · Admin role · Country · Governorate · Notification template · Platform parameter |
| **Tenant foundation** | Branch · User · Role / privilege · Payment method · Parameter · Audit entry |
| **Catalogue** | Clinical group · Sample type · Sample condition · Unit · Service · Service member · Test · Reference range band · List value · Formula · Preparation question · Comment template · Report template · Doctor · Diagnosis · Shortcut set · Test rule · Reason · Patient condition · Sending lab · Organism · Antibiotic · Sensitivity value |
| **Operations** | Patient · Patient document · Visit (accession) · Order line · Preparation answer · Sample · Sample event · Result · Result log entry · Status log entry · Culture result · Text result · Attachment · Print list · Booking · Batch |
| **Finance** | Payer · Contract tier · Base price list · Contract pricing · Price line · Payment · Refund · Transfer · Cancellation · Cash closing · Claim |
| **Integration & automation** | Analyser · Analyser test mapping · Analyser sample code · Site agent · Agent log · PCR kit · Verification settings · Medical rule · Result option · Result dictionary word |

Relationships: a tenant has branches, users, roles and a master-data copy · a patient has visits · a visit has order lines, samples, payments, refunds and documents · a sample has status events and labels · an order line yields results with logs, flags, comments and attachments · a payer has contract tiers whose prices derive from base lists · an analyser has mappings and agent logs · rules belong to tests.

# 10. Documents and reports

| Document | Content | Produced from |
|---|---|---|
| Barcode label | Per tube: unit code, date/time, location, sample type and condition, accession, patient, sample ID barcode, services. | Registration save, Samples by visit |
| Job order | Visit header, balance due, per service unit / expected result / sample / barcode, collection boxes; "collected outside" note. | Registration save, Samples by visit |
| Receipt / invoice | Payment receipt per payment; visit invoice. | Registration (R1), Payments |
| Refund receipt | Amount, method, reason, signature boxes. | Refunds |
| Price quote | Services with shares, expected dates, preparations. | Price query |
| Laboratory report | Template, patient and visit header, results with units, ranges and flags per report unit, comments, verifier signature, QR. | Results, Work queue, Print reports |
| Abbreviated worksheet | Multi-patient file copy with empty result boxes. | Work queue |
| Worklist | Bench list of tests with expected dates. | Worklist |
| Cash-closing sheet | Payments, refunds, expected cash, handover signatures. | Cash closing |
| Claim | Payer visits and shares in the chosen layout with totals and approval boxes. | Claims |
| Analyser code sheet | Mapped codes per analyser. | Analyser tests |
| Financial reports | Sixteen reports with period and branch filters, print and Excel. | Financial reports |

# 11. Non-functional requirements (summary)

| Quality | Requirement |
|---|---|
| Bilingual & RTL | Arabic and English on every screen and document with true right-to-left layout. |
| Responsive | Desktop, tablet and phone; the navigation collapses to a drawer on phones. |
| Security | Tenant isolation, TLS, encrypted health data at rest, OTP channels, password policy, role-based access, device-bound cash, session timeout, audit trail. |
| Performance | 1,500 samples per shift per unit; results screen under 2 s; searches under 3 s; labels within 2 s of save. |
| Availability | 99.9 % monthly target with daily backups and point-in-time recovery. |
| Auditability | Immutable audit records; verification is an electronic signature. |
| Printing | Built-in Code 128 and QR rendering, PDF generation, printer selection per workstation. |
| Configurability | Tenant parameters change behaviour without code changes. |
| Data-entry speed | Keyboard-operable high-volume screens with shortcuts. |
| Migration | Path from LDM for patients, visits, results, catalogue, prices, contracts and users. |

The full non-functional set is in LIS-RS-001, section 4.

# 12. Release scope matrix

## 12.1 Client Portal

| ID | Page | Module | Release 1 | Note |
|---|---|---|---|---|
| CP-01 | **Public website** | Access & onboarding | Yes | |
| CP-02 | **Subscription wizard** | Access & onboarding | Yes | |
| CP-03 | **Login & password recovery** | Access & onboarding | Yes | |
| CP-04 | **Branch selection** | Access & onboarding | Yes | |
| CP-05 | **Dashboard** | Overview | Yes | |
| CP-06 | **Registration** | Reception | Yes | without Location/Department and Room; payment captured before saving |
| CP-07 | **Home reservation** | Reception | Release 2 | |
| CP-08 | **Batch registration** | Reception | Release 2 | |
| CP-09 | **Lab-to-Lab registration** | Reception | Release 2 | |
| CP-10 | **Search registrations** | Reception | Yes | |
| CP-11 | **Patients** | Reception | Yes | |
| CP-12 | **Price query** | Reception | Yes | |
| CP-13 | **Print reports** | Reception | Release 2 | |
| CP-14 | **Collection & receipt** | Samples | Release 2 | in Release 1 samples are received automatically on save |
| CP-15 | **Samples by visit** | Samples | Release 2 | |

| ID | Page | Module | Release 1 | Note |
|---|---|---|---|---|
| CP-16 | **Reject / unreject** | Samples | Release 2 | |
| CP-17 | **Sample tracking** | Samples | Release 2 | |
| CP-18 | **Worklist** | Laboratory | Yes | |
| CP-19 | **Results by accession** | Laboratory | Yes | filters auto-collapse, report units as tabs, no analyser column |
| CP-20 | **Results by sample** | Laboratory | Release 2 | |
| CP-21 | **Work queue** | Laboratory | Release 2 | |
| CP-22 | **Quality control** | Laboratory | Release 2 | |
| CP-23 | **Payments** | Finance | Yes | |
| CP-24 | **Refunds** | Finance | Release 2 | refund action also available on Payments in R1 |
| CP-25 | **Money transfer** | Finance | Release 2 | |
| CP-26 | **Cancellations** | Finance | Release 2 | available as an action on Search registrations in R1 |
| CP-27 | **Batch payment** | Finance | Release 2 | |
| CP-28 | **Treasuries (withdrawn)** | Finance | Release 2 | 2026-10-20, LIS-CR-004: the treasury concept was removed from the system |
| CP-29 | **Treasury requests (withdrawn)** | Finance | Release 2 | 2026-10-20, LIS-CR-004: the treasury concept was removed from the system |
| CP-30 | **Cash closing** | Finance | Release 2 | |
| CP-31 | **Claims** | Finance | Release 2 | |
| CP-32 | **Payers & contracts** | Finance | Yes | |

| ID | Page | Module | Release 1 | Note |
|---|---|---|---|---|
| CP-33 | **Base price lists** | Finance | Yes | |
| CP-34 | **Contract pricing** | Finance | Release 2 | Release 1 uses the seeded contract formulas |
| CP-35 | **Financial reports** | Finance | Release 2 | |
| CP-36 | **Branches** | Setup | Yes | |
| CP-37 | **Roles & privileges** | Setup | Yes | |
| CP-38 | **Users** | Setup | Yes | |
| CP-39 | **Clinical groups** | Setup | Yes | |
| CP-40 | **Services** | Setup | Yes | |
| CP-41 | **Tests** | Setup | Yes | |
| CP-42 | **Service sequence** | Setup | Yes | |
| CP-43 | **Sample types** | Setup | Yes | |
| CP-44 | **Units of measurement** | Setup | Yes | |
| CP-45 | **Patient conditions** | Setup | Release 2 | seeded in Release 1 |
| CP-46 | **Preparations** | Setup | Yes | |
| CP-47 | **Comment library** | Setup | Yes | |
| CP-48 | **Doctors** | Setup | Yes | |
| CP-49 | **Diagnoses** | Setup | Release 2 | seeded in Release 1 |

| ID | Page | Module | Release 1 | Note |
|---|---|---|---|---|
| CP-50 | **Sending labs** | Setup | Release 2 | |
| CP-51 | **Shortcut sets** | Setup | Release 2 | seeded in Release 1 |
| CP-52 | **Test rules** | Setup | Release 2 | seeded in Release 1 |
| CP-53 | **Reasons** | Setup | Release 2 | seeded in Release 1 |
| CP-54 | **Report templates** | Setup | Release 2 | seeded in Release 1 |
| CP-55 | **Additional services** | Setup | Release 2 | seeded in Release 1 |
| CP-56 | **Drug types · Antibiotics · Organisms** | Setup | Yes | |
| CP-57 | **Parameters** | Setup | Release 2 | Release 1 runs on the defaults |
| CP-58 | **Analysers** | Integration | Release 2 | |
| CP-59 | **Analyser tests** | Integration | Release 2 | |
| CP-60 | **Analyser samples** | Integration | Release 2 | |
| CP-61 | **Site agent** | Integration | Release 2 | |
| CP-62 | **PCR kits** | Integration | Release 2 | |
| CP-63 | **Verification settings** | Automation | Release 2 | |
| CP-64 | **Medical rules** | Automation | Release 2 | |
| CP-65 | **Profile** | Account | Yes | |
| CP-66 | **Wallet** | Account | Release 2 | |

| ID | Page | Module | Release 1 | Note |
|---|---|---|---|---|
| CP-67 | **Support** | Account | Release 2 | |
| CP-68 | **Audit log** | Account | Release 2 | |

## 12.2 Super Admin Portal

| ID | Page | Module | Release 1 | Note |
|---|---|---|---|---|
| SA-01 | **Login** | Access | Yes | |
| SA-02 | **Home** | Overview | Yes | |
| SA-03 | **Dashboard** | Overview | Yes | |
| SA-04 | **Clients** | Administration | Yes | |
| SA-05 | **Subscriptions & wallets** | Administration | Yes | |
| SA-06 | **Roles** | Administration | Yes | |
| SA-07 | **Admin users** | Administration | Yes | |
| SA-08 | **Sessions** | Administration | Yes | |
| SA-09 | **Audit log** | Administration | Yes | |
| SA-10 | **Countries** | Master data | Yes | |
| SA-11 | **Governorates** | Master data | Yes | |
| SA-12 | **Clinical groups · Services · Tests · Service sequence · Sample types · Units · Preparations · Comment library** | Master data | Yes | |
| SA-13 | **Patient conditions · Reasons · Diagnoses · Additional services · Report templates · Test rules · Shortcut sets** | Master data | Yes | |
| SA-14 | **Drug types · Commercial names · Antibiotics · Organisms · Sensitivity values** | Microbiology | Yes | |

| ID | Page | Module | Release 1 | Note |
|---|---|---|---|---|
| SA-15 | **Analyser models** | Integration library | Release 2 | |
| SA-16 | **PCR kits** | Integration library | Release 2 | |
| SA-17 | **Code systems** | Integration library | Release 2 | |
| SA-18 | **Rule templates** | Integration library | Release 2 | |
| SA-19 | **Platform parameters** | Platform | Yes | |
| SA-20 | **Notification templates** | Platform | Yes | |
| SA-21 | **Profile** | Platform | Yes | |

# 13. Open questions

| ID | Question |
|---|---|
| OQ-01 | Original Sky LIS SRS referenced by the first prototypes: to be reconciled with this document. |
| OQ-02 | Wallet semantics (prepaid credit, consumption events, behaviour at zero) — modelled here as prepaid credit debited by subscription and metered usage. |
| OQ-03 | Master-data update policy between platform and tenants (push, propose, none). |
| OQ-04 | Whether Release 1 should keep the Received gate (scan-based) instead of auto-receive. |
| OQ-05 | Review/verify order and whether review may be skipped (currently a tenant parameter). |
| OQ-06 | Scope of quality control, outsourcing, SMS and online transfer modules. |
| OQ-07 | Accession and sample ID formats per tenant and label stock sizes. |
| OQ-08 | Analyser protocols and instruments of the pilot laboratory. |

## 13.1 Glossary

| Term | Definition |
|---|---|
| Accession | Visit / order number. |
| Clinical group | Laboratory unit; splits tubes, rights and report sections. |
| Contract tier | Sub-category of a payer with its own price list. |
| Delta check | Comparison with the patient's previous result within a window. |
| Job order | Printout travelling with the tubes. |
| Package | Set of services priced as one. |
| Panic value | Result beyond critical limits requiring immediate notification. |
| Patient condition | Attribute that selects an alternative reference range. |
| Payer | Body paying for services: cash, company, syndicate, insurer, sending lab. |

| Term | Definition |
|---|---|
| Preparation | Question about the patient's readiness asked at registration. |
| Received gate | No result before the sample is received. |
| Report unit | Visit × clinical group (print-alone services separate): what prints as one report. |
| Reserved | Deferred service kept in the visit without a sample. |
| Service | Orderable catalogue item. |
| Turnaround (TAT) | Registration → verification (patient) or receipt → verification (lab). |
| Verification | Doctor's electronic sign-off that releases a result for printing. |
| Work queue | Status-driven list of report units. |